FINISHED PIPE WALL

Measurement and internal-pressure thickness review

ASME B31.1-2022 | Straight pipe | Internal pressure

MINIMUM MEASURED WALL AFTER FABRICATION

3.4787 mm

BELOW the internal-pressure wall requirement

User-entered inspection minimum | Not calculated from Schedule

Required remaining wall	**3.4787 mm**
Measured minus required	**-5.925e-07 mm**

Required remaining wall = pressure thickness + corrosion + erosion. The measured value is the minimum remaining metal after ALL fabrication and machining, including groove/thread roots. It is supplied by the user; the app does not measure the pipe.

No mill under-tolerance or already removed metal is deducted from this measurement again.

01 PIPE AND INSPECTION

Material / grade	**Example steel - verify actual material grade**
Pipe size / outside diameter	**4 inch / 114.3 mm**
Inspection / pipe reference	**مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890**
Project	**Example only - مثال توضيحي**
Prepared by	**QA preview - not a real inspection**
Report date	**2026-10-06 09:07**

02 MEASUREMENT BASIS

Use an inspection minimum that accounts for measurement uncertainty. Exclude coatings, lining and weld reinforcement. This comparison applies to a new straight pipe before service. An average reading or nominal schedule wall is not a minimum finished-wall measurement.

MEETS / BELOW refers only to the internal-pressure wall requirement for these inputs. It is not an inspection certificate or approval of the complete piping design.

DESIGN BASIS

Calculation trail and nominal pipe selection

03 PRESSURE DESIGN INPUTS

Internal design gauge pressure P	**20 bar(g) = 2 MPa**
Design metal temperature	**20 °C**
Allowable stress S	**100 MPa**
Quality factor E / weld factor W	**0.85 / 0.9**
Coefficient Y / y	**0.4**
Material group for y	**Ferritic steels**
Table 104.1.2-1 note (b)	**Not applied**
Corrosion / erosion allowance	**1.5 / 0.5 mm**
Mechanical removal allowance	**1 mm (before machining)**
Code / material input reference	**Illustrative coefficients only; not project design values.**

04 CALCULATION TRACE

P is converted from bar to MPa; D and thicknesses are in mm. S is in MPa.

Pressure thickness t	**P × D / [2 × (S × E × W + P × Y)] = 1.4787 mm**
Required remaining finished wall	**t + corrosion + erosion = 3.4787 mm**
Required wall before machining tm	**t + corrosion + erosion + mechanical = 4.4787 mm**

B31.1-2022 Eq. (7): tm = P Do / [2(SEW + Py)] + A. S is entered before the quality factor; do not apply E twice to a combined SE/SF table stress. The required wall leaves a positive bore.

05 NOMINAL PIPE ORDERING CHECK (SEPARATE)

Specified minus tolerance	**12.5 %**
Minimum nominal pipe wall to order	**tm / (1 - tolerance/100) = 5.1185 mm**
Selected nominal pipe	**Sch 40 / 6.02 mm**
Lower wall bound from tolerance	**5.2675 mm - calculated bound, NOT measured**
Nominal selection result	**Meets the ordering thickness check**

These are pipe wall dimensions. No flat-sheet thickness or forming-process calculation is performed. A nominal-wall check cannot replace the measured finished-wall result at the start of this report.

06 LIMITS AND REFERENCES

Basis: B31.1-2022, 104.1.2(a), equation (7); y: Table 104.1.2-1; source: STRAIGHT PIPE.pdf p.27 and ASME B31.1, 104.pdf p.28. Nominal dimensions: B36.10-2022 Table 2-1. S, E, W, allowances, tolerance and inspection values require verification against project specifications. This report does not assess external pressure, bends, branches, other loads, fatigue, remaining life or high-pressure special services. Comparisons use unrounded numbers; displayed numbers are formatted for readability.